

Cloud Infrastructure Services Proposal
Submitted by: Arctic Cloud Solutions, LLC
In Response to: Fairbanks North Star Borough Enterprise Cloud Infrastructure RFP
Date: July 2026

Executive Summary

Arctic Cloud Solutions (ACS) is pleased to submit this proposal in response to the Fairbanks North Star Borough's Enterprise Cloud Infrastructure RFP. With over 8 years of experience delivering cloud solutions to government entities across Alaska and the Pacific Northwest, we are uniquely positioned to meet the Borough's requirements for a secure, scalable, and compliant cloud platform.

Our proposed solution leverages Microsoft Azure as the primary cloud platform, complemented by our managed services layer providing 24/7 operations, security monitoring, and dedicated Alaska-based support. We have successfully completed similar migrations for the Matanuska-Susitna Borough, the City of Juneau, and the State of Alaska Department of Administration.

Section 1: Vendor Qualifications Response

1.1 Company Background

Arctic Cloud Solutions was founded in 2018 and has provided cloud infrastructure services exclusively to public sector organizations for 8 years. Our team of 45 certified cloud professionals includes 12 based in Fairbanks and Anchorage, ensuring local presence and Alaska business hours coverage.

We currently hold the following certifications:

- SOC 2 Type II (audit completed March 2026, report available upon request)
- ISO 27001:2022
- Microsoft Azure Expert MSP
- AWS Public Sector Partner

Our proposed platform, Microsoft Azure Government, maintains FedRAMP High authorization (not just Moderate), exceeding the Borough's minimum requirement.

References

1. Matanuska-Susitna Borough - Cloud Migration (2023-2024)
Contact: Sarah Chen, IT Director - sarah.chen@matsugov.us - (907) 555-0142
Scope: Full datacenter migration of 200+ workloads to Azure Government
2. City of Juneau - Hybrid Cloud Infrastructure (2022-2023)
Contact: James Whitehorse, CIO - j.whitehorse@juneau.org - (907) 555-0198
Scope: Hybrid cloud deployment with ExpressRoute connectivity
3. State of Alaska, Dept. of Administration - Cloud Security Operations (2024-present)
Contact: Maria Rodriguez, CISO - maria.rodriguez@alaska.gov - (907) 555-0234
Scope: Managed security operations and compliance monitoring

Financial Stability

Arctic Cloud Solutions has demonstrated consistent growth with annual revenues exceeding \$12M in FY2025. Audited financial statements for FY2023, FY2024, and FY2025 are included as Appendix A of this proposal. We maintain a current ratio of 2.3 and zero long-term debt.

Section 2: Project Management Response

Dedicated Project Manager

We will assign Michael Thompson, PMP, CSM, as the dedicated Project Manager for this engagement. Michael is based in Fairbanks and has managed three comparable public sector cloud migrations in Alaska. He is available Monday through Friday during AKST business hours (8:00 AM - 5:00 PM AKST) and carries an after-hours emergency phone.

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50 ### Project Plan
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52 Upon contract execution, our team will deliver a comprehensive project plan within 10
business days (5 days ahead of the required 15 days). The plan will include:
53 - Work breakdown structure with all milestones and deliverables
54 - Resource allocation matrix showing named personnel for each workstream
55 - RACI chart for all project activities
56 - Communication plan aligned with Borough preferences
57 - Quality management plan with acceptance criteria for each deliverable
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59 ### Status Reporting
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61 We commit to:
62 - Weekly status meetings (Tuesday afternoons, AKST) with Borough IT leadership
63 - Written status reports delivered within 24 hours of each meeting
64 - Monthly steering committee presentations with executive dashboard
65 - Real-time project dashboard access via our project management portal
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67 ### Risk Management
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69 Our risk management approach includes:
70 - Risk register maintained in ServiceNow (integrated with Borough's instance)
71 - Monthly risk review at steering committee with mitigation status updates
72 - Proactive risk identification through lessons learned from prior Alaska government
projects
73 - Quantitative risk analysis for high-impact items
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75 ### Escalation Path
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77 Our guaranteed escalation response times:
78 - Critical (P1): 30-minute response, 1-hour engagement of escalation team (exceeds
1-hour requirement)
79 - High (P2): 2-hour response (exceeds 4-hour requirement)
80 - Medium (P3): 4-hour response (exceeds 8-hour requirement)
81 - Low (P4): 48-hour response (best effort)
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83 ## Section 3: Technical Architecture Response
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85 ### 3.1 Cloud Platform
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87 Our solution is built on Microsoft Azure Government, hosted in US sovereign cloud
regions (US Gov Virginia and US Gov Arizona). Key platform capabilities:
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89 ****Hybrid Architecture****: We will deploy Azure Arc to extend Azure management to the
Borough's existing VMware environment, providing a single pane of glass for both
on-premises and cloud workloads. Azure Stack HCI can be deployed on-premises for
workloads requiring local data residency.
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91 ****Availability****: Azure Government provides 99.9% uptime SLA for virtual machines. Our
architecture uses paired regions with manual failover procedures (estimated 45-60
minutes).
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93 ****Operating Systems****: Full support for Windows Server 2019, 2022, and 2025 as well as
Linux distributions including RHEL 8/9, Ubuntu 22.04/24.04, and SUSE Linux Enterprise.
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95 ****Containers****: Azure Kubernetes Service (AKS) with managed control plane, integrated
with Azure Container Registry and Azure Policy for governance.
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97 ****Infrastructure as Code****: Native support for Terraform (with AzureRM provider), Bicep,
and ARM templates. We recommend Bicep for new deployments with Terraform for multi-cloud
scenarios.
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99 ****Serverless****: Azure Functions (consumption and premium plans), Azure Logic Apps, and
Azure Event Grid for event-driven architectures.

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103 ****Managed Databases****: Azure SQL (SQL Server compatible), Azure Database for PostgreSQL Flexible Server, and Azure Cosmos DB (MongoDB API compatible).

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105 **### 3.2 Networking**

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107 ****Private Connectivity****: We will provision a 1 Gbps Azure ExpressRoute circuit from the Borough's primary facility to Azure Government US Gov Virginia, with a 500 Mbps secondary circuit to US Gov Arizona for redundancy.

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109 ****Network Segmentation****: Azure Virtual Networks with Network Security Groups (NSGs) and Azure Firewall provide macro and micro-segmentation. Azure Private Link ensures PaaS services are accessed over private endpoints.

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111 ****DDoS Protection****: Azure DDoS Protection Standard is included for all public-facing endpoints with adaptive tuning and mitigation analytics.

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113 ****IPv4****: Full IPv4 support. IPv6 is not currently supported in our Government cloud deployment.

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115 ****DNS****: Azure DNS for Borough domains. DNSSEC is not available in Azure Government at this time.

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117 ****WAF****: Azure Web Application Firewall with OWASP 3.2 rule set, bot protection, and custom rules for Borough-specific requirements.

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119 ****VPN Backup****: Site-to-site VPN tunnels configured as automatic failover for ExpressRoute. BGP routing ensures seamless failover within 30 seconds.

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123 **### 3.3 Storage**

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125 ****Object Storage****: Azure Blob Storage with 99.999999999% (11 nines) durability in LRS configuration, with GRS providing 16 nines durability for critical data.

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127 ****Tiered Storage****: Hot, Cool, Cold, and Archive tiers with automated lifecycle management policies. We will configure policies aligned with Borough data classification.

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129 ****Block Storage****: Azure Managed Disks with Premium SSD v2 providing configurable IOPS (up to 80,000) and throughput (up to 1,200 MB/s) with SLA-backed guarantees.

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131 ****File Storage****: Azure Files with SMB 2.1 support. NFS is in preview and not available in Government regions.

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133 ****Immutable Storage****: WORM policies for Blob Storage only. Other storage types planned for future.

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135 ****Encryption****: AES-256 encryption at rest for all storage services. Customer-managed keys via Azure Key Vault (FIPS 140-2 Level 3 validated HSM) available for all services.

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137 ****Cross-Region Replication****: GRS and GZRS replication with configurable RPO (typically 15 minutes or less for async replication).

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143 ****Backup and Recovery****: Azure Backup with file-level, volume-level, and application-consistent recovery options. Cross-region restore for disaster recovery scenarios.

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145 **## Section 4: Security Response**

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147 **### 4.1 Identity and Access Management**

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149 ****SSO Integration****: Azure Government natively integrates with Microsoft Entra ID. The
Borough's existing Entra ID tenant will serve as the identity provider for all cloud
resources.

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151 ****MFA****: Microsoft Entra MFA enforced for all administrative access. Supports push
notifications, TOTP, phone call, and hardware security keys.

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153 ****RBAC****: Azure RBAC with 100+ built-in roles. We will implement custom roles following
principle of least privilege, with quarterly access reviews.

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155 ****PAM****: Microsoft Entra Privileged Identity Management (PIM) for just-in-time access
elevation with approval workflows, time-bound assignments, and access reviews.

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157 ****Identity Governance****: Entra ID Governance with access packages, entitlement
management, and quarterly access certification campaigns.

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161 ****Service Accounts****: Managed Identities for Azure resources eliminating credential
management. Where service principals are required, automatic rotation every 90 days.

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163 ****Federation****: B2B collaboration via Entra External ID for inter-agency access without
credential sharing. Trust relationships configurable per partner organization.

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165 ****Audit Logging****: All authentication and authorization events logged to Azure Monitor
and Microsoft Sentinel with tamper-evident storage (immutable log retention).

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169 **### 4.2 Data Protection**

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171 ****TLS****: All data in transit encrypted with TLS 1.3. TLS 1.2 and below disabled across
all services. Certificate pinning for critical service-to-service communication.

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173 ****DLP****: Microsoft Purview DLP policies integrated across Azure services, SharePoint,
Exchange, and Teams. Custom sensitive information types for Borough-specific data
patterns.

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175 ****CMEK****: Customer-managed keys stored in Azure Key Vault with HSM backing (FIPS 140-2
Level 3 via Azure Key Vault Managed HSM).

176
177 ****Data Classification****: Microsoft Purview Information Protection with sensitivity
labels. Auto-classification for common data types with manual override capability.

178
179 ****Key Management****: Azure Key Vault with automatic rotation policies, soft-delete
protection, and purge protection. Dual-person authorization for critical key operations.

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181 ****TDE****: Transparent Data Encryption enabled by default for Azure SQL and other managed
databases. CMEK-backed TDE available for enhanced control.

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185 ****Certificate Management****: Azure Key Vault certificate management with automated
renewal for Let's Encrypt and DigiCert. Integration with Azure Application Gateway for
automated deployment.

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187 ****Secrets Management****: Azure Key Vault for all secrets (API keys, connection strings,
passwords). Full audit logging of all access and operations. Integration with Azure
DevOps for CI/CD pipeline secrets.

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189 ****Data Masking****: Dynamic Data Masking for Azure SQL in production. Static masking via
Microsoft Purview for non-production environment provisioning.

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191 **### 4.3 Threat Protection**

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193 ****SOC****: Our 24/7 Security Operations Center is staffed by certified analysts (CISSP, GCIH, CEH) monitoring Borough environments around the clock. SOC is based in Anchorage with backup facility in Seattle.

194

195 ****SIEM****: Microsoft Sentinel deployed in Azure Government with 180 days of hot storage (exceeds 90-day requirement). Custom analytics rules, workbooks, and hunting queries tailored for government environments.

196

197 ****SOAR****: Sentinel automation rules and playbooks for automated response to common threats including brute force attacks, impossible travel alerts, and known malware indicators.

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199 ****Vulnerability Scanning****: Microsoft Defender Vulnerability Management with daily agent-based scanning and weekly authenticated network scans.

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201 ****EDR****: Microsoft Defender for Endpoint deployed on all virtual machines with automated investigation and response capabilities.

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203 ****Email Security****: Microsoft Defender for Office 365 Plan 2 with anti-phishing, safe attachments, safe links, and attack simulation training.

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205 ****Container Security****: Microsoft Defender for Containers with admission control, runtime protection, and vulnerability scanning integrated into CI/CD pipelines.

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207 ****API Security****: Azure API Management with threat detection. Microsoft Defender for APIs monitoring for anomalous access patterns.

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209 ****Deception Technology****: We deploy custom honeypots within the Borough's cloud environment using Azure Sentinel deception capabilities. Honeytokens placed in Active Directory, key vaults, and storage accounts as early warning indicators.

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211 ****Threat Intelligence****: Microsoft Defender Threat Intelligence integrated with Sentinel. Additional feeds from CISA, MS-ISAC, and Alaska specific threat intelligence sharing communities.

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213 **## Section 5: Compliance and Governance Response**

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215 **### 5.1 Regulatory Compliance**

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217 ****Alaska Data Privacy (AS 45.48)****: Azure Government complies with Alaska state data privacy requirements. All Borough data processed and stored exclusively within US sovereign cloud boundaries.

218

219 ****CJIS****: Azure Government is CJIS compliant. We maintain a signed CJIS Security Addendum and will support the Borough's CJIS audit requirements. Dedicated CJIS-compliant subscription for law enforcement data.

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221 ****HIPAA/HITECH****: Azure Government maintains a signed BAA. HIPAA-compliant configurations pre-deployed for health services workloads with additional safeguards for PHI.

222

223 ****PCI DSS****: Azure Government is PCI DSS Level 1 Service Provider certified. Dedicated PCI-compliant subscription with network isolation for payment processing.

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225 ****ADA Section 508****: Azure Portal and all proposed management interfaces comply with WCAG 2.1 AA standards, meeting Section 508 requirements.

226

227 ****Compliance Dashboards****: Microsoft Defender for Cloud regulatory compliance dashboard with evidence collection. Custom workbooks for Borough-specific compliance frameworks.

228

229 ****Data Sovereignty****: All Borough data remains within Azure Government US regions. Azure Policy enforced to prevent resource creation outside approved regions.

230

231 ****CIS Benchmarks****: Automated assessment against CIS Azure Foundations Benchmark v2.0. Continuous compliance monitoring with automated remediation for common drift.

232

233 ****Records Retention****: Azure Policy enforced retention policies aligned with Borough
records schedule. Immutable storage for records requiring preservation.

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235 ****Chain of Custody****: Full audit trail for all data operations. Azure Activity Log,
Diagnostic Logs, and Storage Analytics provide comprehensive chain of custody
documentation.

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237 **### 5.2 Governance**

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239 ****Resource Tagging****: Azure Policy enforcing mandatory tags (CostCenter, Environment,
Owner, DataClassification, Application, Department). Tag inheritance from resource
groups. Deny deployment if required tags missing.

240

241 ****Policy-as-Code****: Azure Policy definitions managed in Git repository with CI/CD
deployment. Custom policy initiatives for Borough-specific requirements.

242

243 ****Cost Management****: Azure Cost Management + Billing with daily cost reporting, budget
alerts at 80% and 95%, and monthly optimization recommendations. Dedicated FinOps
analyst assigned to Borough account.

244

245 ****Spending Limits****: Azure Budgets with action groups triggering alerts and optional
automated shutdowns for non-production environments.

246

247 ****Change Management****: Azure DevOps pipelines with approval gates for production
deployments. Integration with ServiceNow change management workflows.

248

249 ****Multi-Environment****: Hub-spoke network topology with dedicated subscriptions for
Development, Test, Staging, and Production. Promotion workflows via Azure DevOps with
automated testing gates.

250

251 ****Asset Inventory****: Azure Resource Graph for real-time inventory. Microsoft Defender
for Cloud asset inventory with automatic classification based on data sensitivity.

252

253 ****Sustainability****: Microsoft Emissions Impact Dashboard providing carbon footprint
estimates for all Azure usage. Quarterly sustainability reports included in business
reviews.

254

255 ****License Management****: Azure Hybrid Benefit tracking for Windows Server and SQL Server
licenses. Microsoft 365 license utilization reporting.

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257 ****Organizational Hierarchy****: Azure Management Groups aligned to Borough organizational
structure for cost allocation and policy inheritance.

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259 **## Section 6: Operations and Support Response**

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261 **### 6.1 Monitoring and Observability**

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263 ****Comprehensive Monitoring****: Azure Monitor covering infrastructure metrics, application
insights, and custom business metrics. Single-pane dashboard for Borough IT leadership.

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265 ****Custom Dashboards****: Azure Dashboards and Grafana integration with real-time
visualization. Configurable refresh intervals from 5 seconds to 24 hours.

266

267 ****Alerting****: Azure Monitor Alerts with action groups supporting email, SMS, Microsoft
Teams, PagerDuty, and ServiceNow ticket creation.

268

269 ****Distributed Tracing****: Application Insights with end-to-end distributed tracing for
microservices. OpenTelemetry support for custom instrumentation.

270

271 ****Log Aggregation****: Log Analytics workspace with structured KQL queries. 1-year hot
retention with 7-year cold retention for compliance requirements.

272

273 ****Synthetic Monitoring****: Azure Monitor availability tests with multi-step web tests
from multiple global locations (US-focused for Borough services).

274

275 ****Capacity Planning****: Azure Advisor recommendations with predictive analytics via

AI-powered insights. Monthly capacity reviews in operational meetings.

276

277 ****Custom Metrics****: Application Insights custom metrics API and Azure Monitor custom metrics for Borough-developed applications.

278

279 ****SLO Tracking****: Custom SLO dashboards in Grafana with error budget burn rate alerts. Monthly SLO reports included in operational reviews.

280

281 ****AIOps****: Azure Monitor AIOps with smart detection, anomaly grouping, and root cause analysis. Reduces alert noise by correlating related issues.

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283 ### 6.2 Disaster Recovery

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285 ****RTO - Tier 1****: Automated failover for Tier 1 applications with 2-hour RTO (exceeds 4-hour requirement). Azure Site Recovery with continuous replication.

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287 ****RPO - Tier 1****: 15-minute RPO for Tier 1 applications using synchronous replication for databases and near-sync for application tiers.

288

289 ****DR Testing****: Monthly automated DR failover tests in isolated recovery environment. No impact to production workloads. Automated validation scripts confirm application functionality.

290

291 ****Runbook Automation****: Azure Automation runbooks for all DR procedures. Regularly tested and updated with each infrastructure change.

292

293 ****DR Reports****: Quarterly DR drill reports with detailed findings, remediation tracking, and improvement recommendations. Shared at steering committee.

294

295 ****Backup Immutability****: Azure Backup with immutable vaults. Multi-user authorization required for modifying backup policies. Ransomware protection validated quarterly.

296

297 ****Cross-Region DR****: Active-passive configuration between US Gov Virginia (primary) and US Gov Arizona (secondary). Per-application RPO/RTO configuration.

298

299 ****Application-Consistent Snapshots****: Azure Backup with VSS integration for Windows and custom scripts for Linux. Database-aware snapshots for SQL Server, PostgreSQL.

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301 ****DR Orchestration****: Azure Site Recovery Plans with dependency-aware sequencing. Custom pre/post scripts for application-specific recovery steps.

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305 ### 6.3 Support

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307 ****24/7 Support****: Follow-the-sun support model with primary team in Anchorage (AKST hours) and backup teams in Seattle and Atlanta for after-hours coverage.

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309 ****TAM****: Dedicated Technical Account Manager based in Fairbanks with deep knowledge of Borough infrastructure. Weekly office hours and monthly on-site visits.

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312

313 ****Training****: 30-hour initial training covering Azure fundamentals. 8 hours annual online-only refresher.

314

315 ****Business Reviews****: Quarterly business reviews with capacity planning, cost optimization, security posture assessment, and roadmap alignment.

316

317 ****Roadmap****: 6-month service roadmap updated semi-annually. 30-day advance notice for breaking changes.

318

319 ****Post-Mortem****: Incident post-mortem reports delivered within 3 business days (exceeds 5-day requirement) for P1/P2 incidents. Blameless format with systemic improvements.

320

321 ****ServiceNow Integration****: Bi-directional integration with Borough's ServiceNow

instance. Automated ticket creation, status sync, and resolution updates.

****Dedicated Support**:** Named support team of 4 engineers dedicated to the Borough account. Direct Slack/Teams channel for real-time communication.

****Professional Services**:** Pre-negotiated rates for architecture reviews (\$175/hr), migrations (\$150/hr), and optimization projects (\$165/hr). 500 hours per year included in base contract.

Section 7: Migration Response

7.1 Migration Planning

****Discovery**:** We will deploy Azure Migrate appliances within 2 weeks of contract start, completing full discovery and assessment within 25 days (ahead of 30-day requirement). Assessment covers compute, storage, network, and application dependencies.

****Migration Strategy**:** Each workload categorized using the 6 Rs framework (Rehost, Replatform, Refactor, Repurchase, Retire, Retain). Recommendation based on TCO analysis, risk assessment, and Borough priorities.

****Zero-Downtime**:** For critical applications, we employ blue-green deployment patterns with Azure Traffic Manager for seamless cutover. Database migrations use online migration tools (DMS) for continuous sync.

****Rollback**:** Every migration wave includes documented rollback procedures tested in DR environment before production execution. Recovery validated within 2 hours.

****Parallel Run**:** Minimum 30-day parallel run for all Tier 1 and Tier 2 applications. Extended to 60 days if requested by Borough application owners.

7.2 Data Migration

****Secure Transfer**:** AzCopy with AES-256 encryption in transit and at rest. SHA-256 checksums validated pre and post transfer. Transfer logs retained for audit.

****Database Migration**:** Azure Database Migration Service with schema conversion for incompatible sources. Pre-migration compatibility assessment and remediation guidance.

****Data Validation**:** Automated validation framework comparing source and destination row counts, checksums, and sample data verification. 100% integrity certification signed off by Borough data owners.

****Progress Tracking**:** Daily migration status reports via email and dashboard. Real-time progress tracking in Azure Migrate portal accessible to Borough team.

****Offline Transfer**:** Azure Data Box (100 TB) and Data Box Heavy (1 PB) available for large datasets. NIST-compliant encryption with tamper-evident shipping. Chain of custody documentation provided.

Section 8: Financial Response

8.1 Pricing

****Transparency**:** All pricing detailed in Appendix C with no hidden fees. Egress costs, support fees, and professional services rates clearly itemized.

****Committed Use**:** 1-year commitment: 20% discount. 3-year commitment: 35% discount. Discounts applied to all Azure consumption and managed services fees.

****Cost Model**:** Detailed cost model for Borough projected workload (Appendix B) included in Appendix C. Monthly estimated cost: \$47,500 (1-year term) or \$39,200 (3-year term).

****Price Cap**:** Annual increases capped at 8% for the duration of the contract.

****Pilot Period**:** 45-day trial with up to \$15,000 in credits. Contract commitment

required after trial.